Kruskal's Algorithm-

- Kruskal's Algorithm is a famous greedy algorithm.
- It is used for finding the Minimum Spanning Tree (MST) of a given graph.
- To apply Kruskal's algorithm, the given graph must be weighted, connected and undirected.

Kruskal's Algorithm Implementation-

Step-01:

- Sort all the edges from low weight to high weight.

Step-02:

- Take the edge with the lowest weight and use it to connect the vertices of graph.
- If adding an edge creates a cycle, then reject that edge and go for the next least weight edge.

Step-03:

- Keep adding edges until all the vertices are connected and a Minimum Spanning Tree (MST) is obtained.

Kruskal's Algorithm Time Complexity-

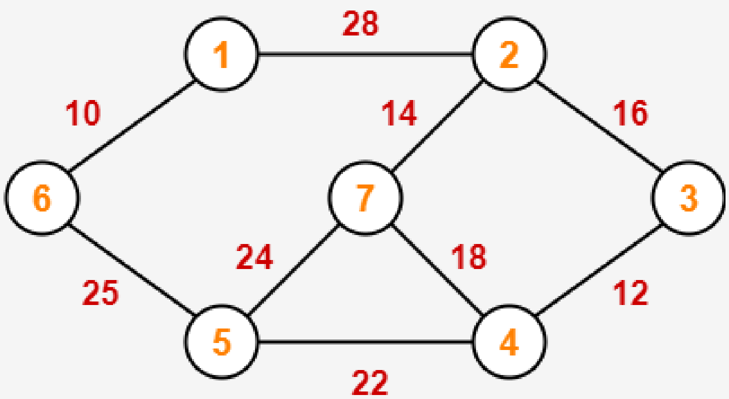
Worst case time complexity of Kruskal's Algorithm = O(ElogV) or O(ElogE)

If the edges are already sorted,

Best Case time complexity of Kruskal's Algorithm = O(E + V)

Problem-01:

Construct the minimum spanning tree (MST) for the given graph using Kruskal's Algorithm-

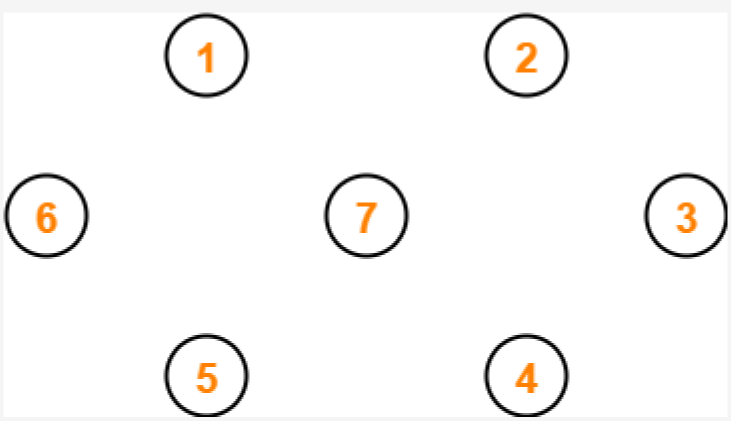


Solution-

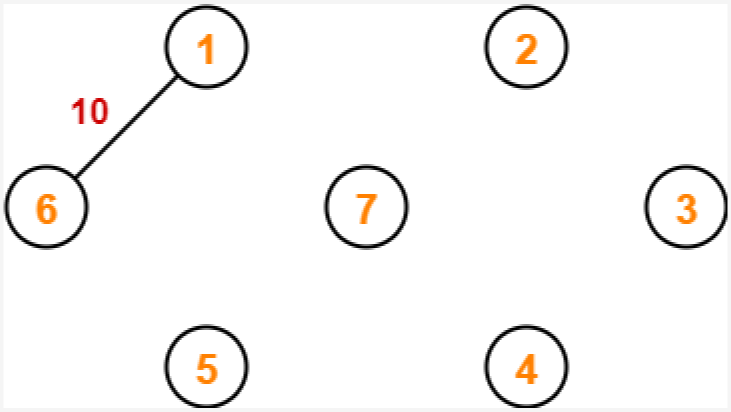
To construct MST using Kruskal's Algorithm,

- Simply draw all the vertices on the paper.
- Connect these vertices using edges with minimum weights such that no cycle gets formed.

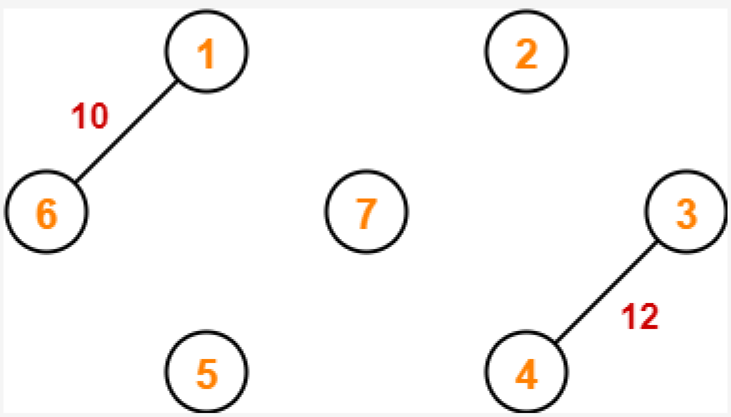
Step-01:



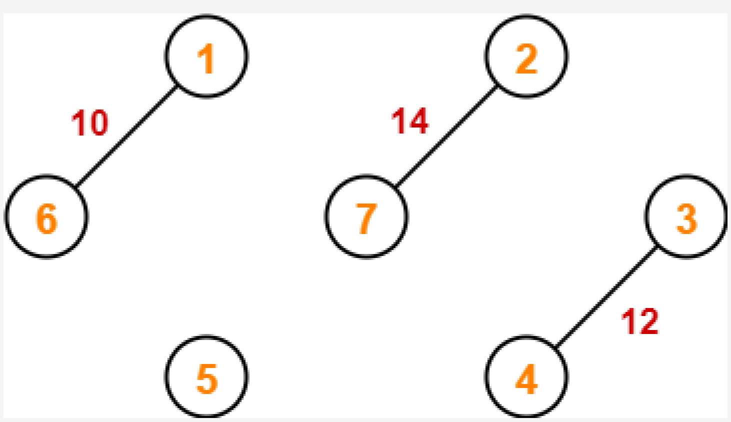
Step-02:



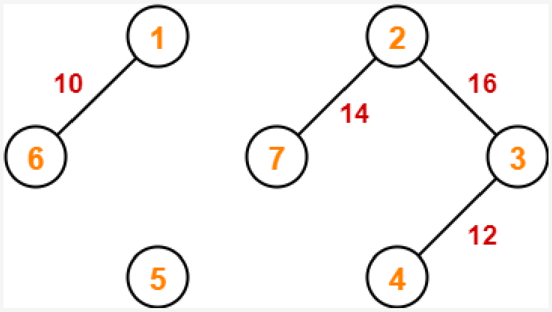
Step-03:



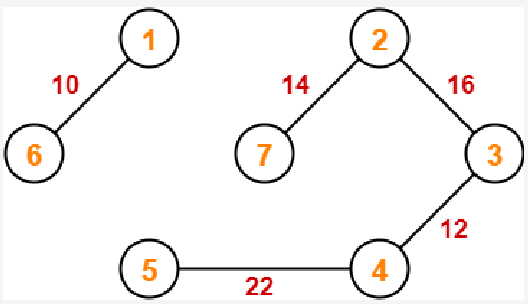
Step-04:



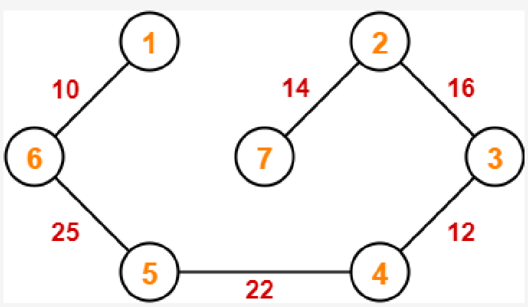
Step-05:



Step-06:



Step-07:



Since all the vertices have been connected / included in the MST, so we stop.

Weight of the MST

= Sum of all edge weights

= 10 + 25 + 22 + 12 + 16 + 14= 99 units